# TWO ENDOMORPHISMS OF A TRIANGLE COMPOSE TO ZERO ON THE MIDDLE TERM

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 710-728, following the statement at 700-708.

*Editorial note.* No omissions: the diagram, the dotted arrows and the vanishing are all displayed or named.

*Allowed standard prerequisites (editorial).*

- Representable functors are homological. lemma-representable-homological (carded L26235, cited at ~721). Supplies the dotted arrows alpha and beta.
- Composition of consecutive triangle maps is zero. lemma-composition-zero (carded L26252, cited at ~725). g o f = 0 kills the composite.

*Source reference.* Original lemma 0146: sources/stacks-derived/pinned-original-source.tex, source lines 575–575. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

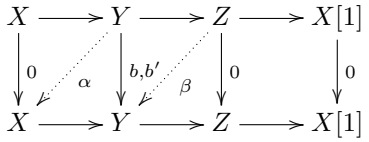
*Source reference.* Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $\mathcal{D}$ be a pre-triangulated category. Let*

$$(0, b, 0), (0, b', 0) : (X, Y, Z, f, g, h) \to (X, Y, Z, f, g, h)$$

*be endomorphisms of a distinguished triangle. Then $bb' = 0$.*

*Proof.* Picture




Applying Lemma 0149 we find dotted arrows $\alpha$ and $\beta$ such that $b' = f \circ \alpha$ and $b = \beta \circ g$. Then $bb' = \beta \circ g \circ f \circ \alpha = 0$ as $g \circ f = 0$ by Lemma 0146. □

*Date*: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.